

Framing the Fake: The Differential Effect of Forgery vs. AI Labels on Aesthetic Experience

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TODO

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- Framing the Fake: The Psychological Impact of Forgery and AI Beliefs on Aesthetic Experience
- Art-ificial Beauty
- The Ghost in the Canvas

Introduction

Methods

Participants

TODO.

Procedure

Artwork Judgment Task

Participants were first introduced to a cover story framing the experiment as a collaboration between neuroscientists and artists investigating how people visually explore and evaluate artworks (a framing that also justified the use of webcam-based eye-tracking). They were told they would view three categories of images: Original Human paintings (sourced from public art databases), AI-Generated images (produced using generative platforms such as Midjourney and Stable Diffusion, either in a new style or inspired by existing artists or artworks¹), and Human Forgeries (copies or stylistic imitations of existing artworks, historically produced by Humans with intent to deceive). Critically, only the participants' *beliefs* were manipulated - all stimuli were, in reality, genuine man-made artworks.

Stimuli from the Vienna Art Picture System (VAPS, Fekete et al., 2023) were first grouped into four broad stylistic categories (Abstract and Avant-garde; Impressionist and Expressionist; Romantic and Realism; and Classical). To minimize the influence of prior exposure, only unfamiliar paintings (familiarity ratings below the sample median) were retained. Within each style, the remaining paintings were stratified into four quadrants defined by the upper and lower tertiles of normative arousal and valence, and the three paintings farthest from the style's median arousal/valence were selected from each quadrant. This procedure yielded a final set of 48 paintings - 12 per style, evenly split across the four arousal-valence quadrants (Positive - High Intensity; Positive - Low Intensity; Negative - High Intensity, Negative - Low Intensity).

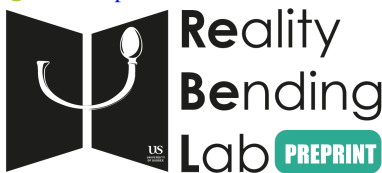
¹This intentionally equivocal description, that leaves open whether AI images were generated with the goal of being new or artist-inspired, was carefully considered and included to preempt the risk that a participant recognizing a genuine painting presented as AI-generated would undermine belief in the whole manipulation.

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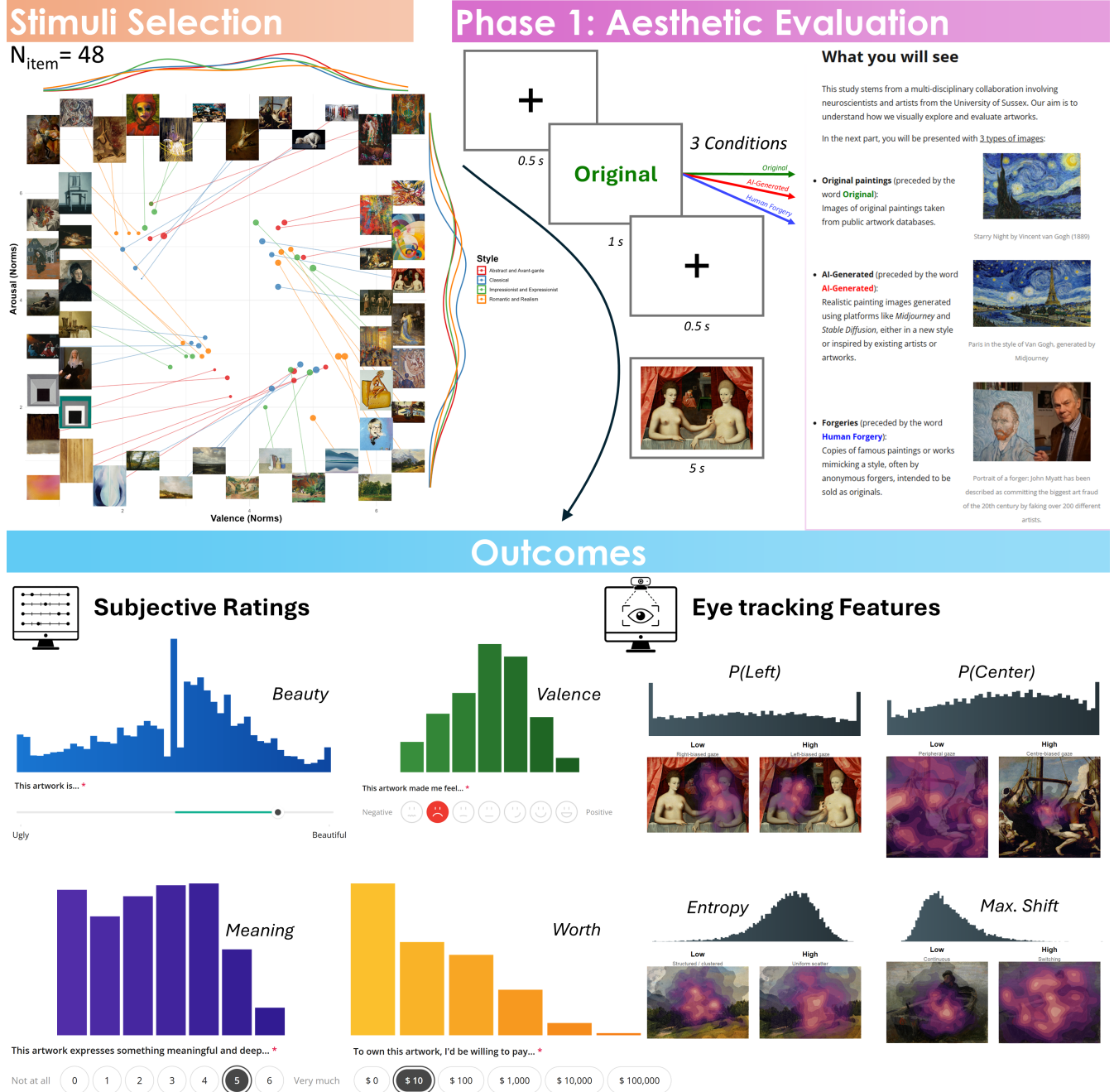


Author roles were classified using the Contributor Role Taxonomy (CRediT; <https://credit.niso.org/>) as follows: Dominique Makowski: Project administration, Data curation, Formal Analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing; Ana Neves: Project administration, Writing – review & editing; Marco Sperduti: Project administration, Writing – original draft, Writing – review & editing

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Figure 1

TODO.



41 During the first phase of the experiment, participants
 42 viewed the 48 artworks (see Figure 1), each preceded by
 43 a randomly presented label indicating its purported type
 44 ("Original", "Human Forgery" or "AI-Generated"). Follow-
 45 ing the presentation of each artwork (5 s), participants re-
 46 ported their aesthetic experience across four dimensions en-
 47 compassing cognitive appraisal, emotional response, seman-

48 tic processing, and valuation (Kirk et al., 2009; Leder et
 49 al., 2004; Leder & Nadal, 2014): *beauty* (aesthetic quality,
 50 composition, and execution), emotional *valence* (the inten-
 51 sity of positive or negative feelings evoked), *meaningfulness*
 52 (conceptual richness, symbolism, and profundity), and mon-
 53 etary *worth* (maximum willingness to pay). To mitigate com-
 54 mon method bias (Podsakoff et al., 2003) and minimize con-

struct overlap, each dimension was assessed with a bespoke scale: an analog slider (Ugly - Beautiful), a 7-point pictorial scale (Negative - Positive), a 7-point numerical scale from 0 (Not at all) to 6 (Very much), and a 6-point discrete logarithmic monetary scale offering choices from \$0 to \$100,000². To ensure sustained engagement with the contextual cues, random catch trials were interleaved throughout the block, prompting participants to recall the immediately preceding label.

At the start of the second phase, the cover story was *partially* revealed³. Participants were informed that some of the Phase 1 labels had been intentionally randomized and were tasked with identifying the true nature of the artworks based on their own judgment (preserving the belief that originals, forgeries and AI-generated stimuli were present). For each briefly re-presented image (1.5 s), participants made two continuous, gut-feeling judgments regarding the artwork's nature: *syntheticness* ("AI-generated" vs. "Human Creation") and *authenticity* ("Copy/forgery" vs. "Original Creation"). This two-phase design allowed us to effectively isolate the top-down influence of semantic labels on aesthetic experience from the bottom-up perceptual detection of synthetic or forged art.

Questionnaires

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Memory

Participants were invited after $\pm X$ days to a follow-up session [Explain follow-up procedure and memory test].

Data Analysis

Preprocessing

Outlier rejection.

Eye Tracking

Gaze position was sampled continuously via webcam-based eye-tracking (WebGazer.js) throughout each trial and re-expressed in coordinates centered on the stimulus and normalized by stimulus width, preserving the image's aspect ratio and allowing comparison across images of different sizes. Given the noisy and variable nature of webcam-based data, a strict quality control workflow was applied. X participants were excluded due to invalid data ($>25\%$ consecutive duplicate samples). Off-screen coordinates and samples reflecting implausibly fast transitions (>6 stimulus-widths/second, indicative of blinks or artifacts) were removed. Finally, trials were excluded if they contained fewer than 20 valid samples, showed frozen ($SD < 0.02$) or excessively noisy ($SD > 0.60$) gaze on either axis, had less than 50% of samples falling within the stimulus bounds, exhibited severe linear drift (>1 stimulus-width of total displacement), or showed

near-uniform spatial scatter consistent with untracked gaze. Finally, X participants were excluded for yielding fewer than 2,000 valid samples across the full experiment.

Four simple, robust and interpretable gaze features were computed per trial. *Laterality* was the proportion of samples falling in the left half of the image (vs. the right), and *Centeredness* was the proportion of samples falling within the center portion of the image. *Entropy* quantified the spatial dispersion of gaze: samples were binned into a 5×5 grid, Shannon entropy was computed and normalized by the maximum entropy achievable given the number of valid samples (avoiding artificial deflation on sparse trials). *Max. Shift* captured the single largest reorganization of gaze focus within a trial: for every possible partition of the chronologically ordered samples into an initial and final segment (each with at least 30 samples), the Euclidean distance between the two segments' centroids was computed, and the maximum across all partitions was retained; large values indicate a discrete relocation of attention to a different image region part-way through viewing, near-zero values indicate a single stable focus throughout.

Statistical Models

- First step: Linear Mixed Models. apply linear mixed effect models to iterate through a number of various features and comprehensively assess important and relevant predictors.
- However, while linear models model the expected mean of the outcome, it is to note that they are not the model class the most suited for our data. On a theoretical level, the outcomes were recorded on scales with particular characteristics (bounded at extremities, some had discrete choices, etc.). Critically, at an empirical level, most display distributions that are better accounted for by other models. For instance, Meaning and Worth displayed characteristic zero-inflation, had Beauty, Syntheticness and Authenticity displayed bimodal distributions with zero, one and mid-point inflations. The latter type of distributions have been recently described as a Choice-Confidence distribution, where the response reflect a conflated process of a discrete choice - e.g., Beautiful vs. Ugly - and a continuous evaluation of degree or confidence. These distributions are better modeled using the Choice-Confidence models, a mixture model of two Beta distributions that provides a much better fit than linear

²The selection, description, wording and presentation of these measures were refined through extensive pilot testing and expert feedback.

³This pseudo-reveal paradigm was developed as an improvement over other designs in related experiments, see **FICTIONERO PREPRINT REF** for discussion

models as well as deeper insights, with parameters reflecting the probability of each discrete choice, as well as the mean and spread of each choice separately.

Discussion

While empirical aesthetics research frequently observes high correlations among subjective dimensions - often prompting the extraction of a single latent “aesthetic appreciation” factor to mitigate multicollinearity, the present data strongly support preserving the independence of these constructs. Theoretically, beauty, valence, meaning, and worth map onto distinct cognitive and neural substrates, spanning sensory evaluation, affective response, semantic processing, and reward. Crucially, the empirical distributions of the collected data substantiate this theoretical divergence. Rather than sharing a uniform shape, each dimension exhibited a distinct distributional signature indicative of separate underlying generative processes. Specifically, emotional valence displayed a symmetric unimodal distribution, reflecting a normative, continuous gradient of affective responses. In contrast, meaningfulness demonstrated pronounced zero-inflation, suggesting a partially dichotomous cognitive mechanism whereby artworks are initially gated as either possessing semantic weight or lacking it entirely. Monetary worth followed a monotonically decaying, right-skewed shape highly characteristic of behavioral economic valuation paradigms, where high subjective reward is exceptionally rare. Finally, beauty judgments yielded a complex, bimodal distribution, likely capturing a combination of po-

larized aesthetic processing and the specific anchoring behaviors elicited by analog sliders. These pronounced morphological differences, while helped by our usage of distinct response formats, suggest that the four scales are not merely redundant probes of a generalized aesthetic halo effect, but capture distinct aspect of the aesthetic experience.

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